Solutions for Real Analysis HW 3

Problem 1. Is the function $f(x,y) = e^{-xy}\sin x \sin y$ integrable on $[0,+\infty)\times[0,+\infty)$ w.r.t. the standard Lebesgue measure?

Solution. By Tonelli's Theorem (the inverse Fubini Theorem, Problem 2), it suffices to show that the iterated integral of $|f|$ is finite:

$$I = \int_0^\infty |\sin y| \left(\int_0^\infty e^{-xy} |\sin x| \, dx \right) dy.$$

We split the outer integral into two parts: $[0,1]$ and $(1,+\infty)$.

1. **On** $(0,1]$**:** Using the bound $|\sin x| \le 1$, the inner integral satisfies

$$\int_0^\infty e^{-xy}|\sin x| \, dx \le \int_0^\infty e^{-xy} \, dx = \frac{1}{y}.$$

   Thus, the integrand is bounded by $|\sin y| \cdot \frac{1}{y} \le \frac{y}{y} = 1$, which is integrable on $[0,1]$.

2. **On** $(1,+\infty)$**:** Using the bound $|\sin x| \le x$, the inner integral satisfies

$$\int_0^\infty e^{-xy}|\sin x| \, dx \le \int_0^\infty x e^{-xy} \, dx = \frac{1}{y^2}.$$

   Thus, the integrand is bounded by $|\sin y| \cdot \frac{1}{y^2} \le \frac{1}{y^2}$, which is integrable on $(1,+\infty)$.

Since the integral converges on both intervals, f is integrable. □

Problem 2. Show the validity of Fubini theorem for nonnegative measurable functions on a measurable set A w.r.t. the product measure without assuming $f \in L_1(A)$.

Solution. This is the Tonelli Theorem. Let $f(x,y) \ge 0$ be measurable. Construct a sequence $f_n(x,y) = \min(f(x,y),n) \cdot \chi_{A\cap B_n(0)}$, where $B_n(0)$ is the ball of radius n. Each f_n is bounded and supported on a set of finite measure, hence $f_n \in L_1$. Fubini's theorem holds for f_n:

$$\int_Y \int_X f_n \, d\mu_x d\mu_y = \int_{X\times Y} f_n \, d(\mu_x \times \mu_y) = \int_X \int_Y f_n \, d\mu_y d\mu_x.$$

Since $0 \le f_n \nearrow f$, by the Levi Convergence Theorem, the integrals converge to the integrals of f (which may be $+\infty$). Thus the equality holds for f. □

Problem 3. Show the failure of Fubini theorem for the standard Lebesgue measure on $\mathbb{R}^n$, the set $A = [-1,1]\times[-1,1]$, and $f(x,y) = xy/(x^2+y^2)^2$.

Solution. **Iterated Integrals:** For fixed $y \neq 0$, the function $x \mapsto \frac{xy}{(x^2+y^2)^2}$ is odd on $[-1,1]$. Thus $\int_{-1}^{1} f(x,y)dx = 0$. Integrating this w.r.t y gives 0. Similarly, integrating w.r.t y first gives 0. So both iterated integrals exist and equal 0.

Double Integral (Integrability): We check if $f \in L_1(A)$. Consider polar coordinates.

$$\int_A |f| \geq \int_D \frac{|r^2 \cos\theta \sin\theta|}{r^4} r dr d\theta = \int_0^{2\pi} |\cos\theta \sin\theta| d\theta \int_0^1 \frac{1}{r} dr.$$

The radial integral $\int_0^1 \frac{1}{r} dr = [\ln r]_0^1 = \infty$. Since $\int_A |f| = \infty$, the function is not Lebesgue integrable on A. The assumption of Fubini's theorem ($f \in L_1$) is violated, which is why the iterated integrals can exist (and equal) while the double integral is undefined. □

Problem 4. Using Lebesgue integration theory, show that the polar substitution is valid for $f \in C(D)$.

Solution. Let $\Phi(r,\theta) = (r\cos\theta, r\sin\theta)$. Since f is continuous on the compact set D, it is bounded. Let $|f(x,y)| \leq M$.

To rigorously apply the Change of Variables Theorem for the Riemann integral, the mapping must be a diffeomorphism (bijective and differentiable with non-zero Jacobian) on the domain. The map Φ fails to be injective on the boundary of the annulus (identifying $\theta = 0$ and $\theta = 2\pi$) and has a vanishing Jacobian at $r = 0$. Therefore, we approximate the domain by cutting out the origin and a small angular sector.

Consider the sequence of compact rectangles in the (r,θ) plane for $n \in \mathbb{N}$ (large enough):

$$K_n = [1/n, 1] \times [1/n, 2\pi - 1/n].$$

Let $E_n = \Phi(K_n) \subset D$. On the domain K_n, the map Φ is strictly injective, and the Jacobian determinant $J(r,\theta) = r \geq 1/n > 0$. Thus, Φ is a diffeomorphism from K_n onto E_n, and the standard Riemann Change of Variables formula applies:

$$\int_{E_n} f(x,y)\, dxdy = \int_{K_n} f(r\cos\theta, r\sin\theta) \cdot r\, drd\theta.$$

Now we take the limit as $n \to \infty$.

1. **LHS:** Let χ_{E_n} be the characteristic function of E_n. As $n \to \infty$, the sets E_n expand to fill D except for the origin and the line segment corresponding to $\theta = 0$. Since these excluded sets have Lebesgue measure zero, $\chi_{E_n}(x,y) \to \chi_D(x,y)$ almost everywhere. Since $|f(x,y)\chi_{E_n}(x,y)| \leq M\chi_D(x,y)$ and D has finite measure, by the Lebesgue Convergence Theorem:
$$\lim_{n\to\infty} \int_{E_n} f(x,y)\, dxdy = \int_D f(x,y)\, dxdy.$$

2. **RHS:** Similarly, let $g(r,\theta) = f(r\cos\theta, r\sin\theta)r$. The sequence of sets K_n expands to $R = [0,1] \times [0,2\pi]$. Thus, $\chi_{K_n} \to \chi_R$ almost everywhere. Since g is bounded on R, by the Dominated Convergence Theorem again:
$$\lim_{n\to\infty} \int_{K_n} g(r,\theta)\, drd\theta = \int_0^{2\pi} \int_0^1 f(r\cos\theta, r\sin\theta) r\, drd\theta.$$

Equating the limits, we obtain the polar integration formula:

$$\int_D f(x,y)\, dxdy = \int_0^{2\pi} d\theta \int_0^1 f(r\cos\theta, r\sin\theta) r\, dr.$$

□

Problem 5. Prove that Hölder inequality turns into an equality iff $a|f|^p = b|g|^q$.

Solution. Hölder's inequality is derived from the following elementary inequality: for any non-negative real numbers $u, v \geq 0$ and conjugate exponents p, q (i.e., $1/p + 1/q = 1$),

$$uv \leq \frac{u^p}{p} + \frac{v^q}{q},$$

with equality if and only if $u^p = v^q$.

To prove Hölder's inequality for functions f and g, we assume $\|f\|_p > 0$ and $\|g\|_q > 0$ (otherwise the result is trivial). We apply the inequality above pointwise with $u = \frac{|f(x)|}{\|f\|_p}$ and $v = \frac{|g(x)|}{\|g\|_q}$:

$$\frac{|f(x)g(x)|}{\|f\|_p\|g\|_q} \leq \frac{1}{p}\frac{|f(x)|^p}{\|f\|_p^p} + \frac{1}{q}\frac{|g(x)|^q}{\|g\|_q^q}.$$

Integrating both sides over the space yields:

$$\frac{\int |fg|}{\|f\|_p\|g\|_q} \leq \frac{1}{p}\underbrace{\int \frac{|f|^p}{\|f\|_p^p}}_{=1} + \frac{1}{q}\underbrace{\int \frac{|g|^q}{\|g\|_q^q}}_{=1} = \frac{1}{p} + \frac{1}{q} = 1.$$

Thus, $\int |fg| \leq \|f\|_p\|g\|_q$. Equality in the integral holds if and only if the pointwise inequality becomes an equality almost everywhere. According to the condition $u^p = v^q$, equality holds iff:

$$\left(\frac{|f(x)|}{\|f\|_p}\right)^p = \left(\frac{|g(x)|}{\|g\|_q}\right)^q \quad \text{a.e.}$$

Rearranging terms, this is equivalent to $\|g\|_q^q \cdot |f|^p = \|f\|_p^p \cdot |g|^q$ a.e., which satisfies the form $a|f|^p = b|g|^q$ with constants $a = \|g\|_q^q$ and $b = \|f\|_p^p$. □

Problem 6. Find a criterion for $L_p \subset L_r$ on (a) $(0, 1)$ and (b) $(0, +\infty)$.

Solution. **(a) On** $(0, 1)$**:** $L_p(0, 1) \subseteq L_r(0, 1)$ if and only if $p \geq r$.

- *Proof ($p \geq r$):* Since the measure of the domain is finite ($\mu = 1$), by Hölder's inequality: $\|f\|_r \leq \|f\|_p \cdot \|1\|_{pr/(p-r)} < \infty$.
- *Counterexample ($p < r$):* Consider $f(x) = x^{-\alpha}$. $f \in L_p \iff \int x^{-\alpha p} < \infty \iff \alpha p < 1$. $f \in L_r \iff \alpha r < 1$. If $p < r$, we can choose α such that $\frac{1}{r} < \alpha < \frac{1}{p}$. Then $f \in L_p$ but $f \notin L_r$.

(b) On $(0, +\infty)$**:** $L_p(0, \infty) \subseteq L_r(0, \infty)$ if and only if $p = r$.

- *Proof ($p = r$):* Trivial.
- *Counterexample ($p \neq r$):* If $p < r$: Use the same function as above near 0, e.g., $f(x) = x^{-\alpha}\chi_{(0,1)}$ with $1/r < \alpha < 1/p$. Then $f \in L_p \setminus L_r$. If $p > r$: We need a function that decays slowly at infinity. Consider $f(x) = x^{-\beta}\chi_{(1,\infty)}$. $f \in L_p \iff \beta p > 1$. $f \in L_r \iff \beta r > 1$. Choose β such that $\frac{1}{p} < \beta < \frac{1}{r}$. Then $f \in L_p$ but $f \notin L_r$.

□

Problem 7. Let $1 \leq p < q < r < +\infty$. Show that $L_q(\mathbb{R}) \subset L_p(\mathbb{R}) + L_r(\mathbb{R})$.

Solution. Let $f \in L_q(\mathbb{R})$. We decompose f based on the magnitude of its values. Define:

$$g(x) = f(x) \cdot \mathbf{1}_{\{x:|f(x)|>1\}}, \quad h(x) = f(x) \cdot \mathbf{1}_{\{x:|f(x)|\leq 1\}}.$$

Clearly $f = g + h$.

1. **Check $g \in L_p$:** On the set where $|f| > 1$, we have $|f|^p < |f|^q$ (since $p < q$).

$$\int |g|^p = \int_{\{|f|>1\}} |f|^p \leq \int_{\{|f|>1\}} |f|^q \leq \|f\|_q^q < \infty.$$

   Thus $g \in L_p(\mathbb{R})$.

2. **Check $h \in L_r$:** On the set where $|f| \leq 1$, we have $|f|^r \leq |f|^q$ (since $r > q$).

$$\int |h|^r = \int_{\{|f|\leq 1\}} |f|^r \leq \int_{\{|f|\leq 1\}} |f|^q \leq \|f\|_q^q < \infty.$$

   Thus $h \in L_r(\mathbb{R})$.

Therefore, $f \in L_p + L_r$. □

Problem 8. Let $f_n \to f$ in measure, $|f_n| \leq F \in L_p$. Prove $f_n \to f$ in L_p.

Solution. Since $|f_n| \leq F$ and $f_n \to f$ in measure, we have $|f| \leq F$ a.e., and thus $|f_n - f|^p \leq (2F)^p \in L_1$. For any $\epsilon > 0$:

1. **Control the "tail" (small values):** Consider the function $h_\delta = \min\{\delta^p, (2F)^p\}$. Since $h_\delta \searrow 0$ a.e. as $\delta \to 0$ and $h_\delta \leq (2F)^p \in L_1$, by the Lebesgue Convergence Theorem, there exists $\delta > 0$ such that $\int_\Omega h_\delta \, d\mu < \epsilon/2$.

2. **Control the "peak" (small measure):** Since $(2F)^p \in L_1$, by the absolute continuity of the Lebesgue integral, there exists $\eta > 0$ such that for any measurable set E with $\mu(E) < \eta$, we have $\int_E (2F)^p \, d\mu < \epsilon/2$.

3. **Combine:** By convergence in measure, for the fixed δ, there exists N such that for all $n > N$, $\mu(E_n) < \eta$, where $E_n = \{x : |f_n(x) - f(x)| \geq \delta\}$. Split the integral:

$$\begin{aligned}
\int_\Omega |f_n - f|^p d\mu &= \int_{E_n} |f_n - f|^p d\mu + \int_{E_n^c} |f_n - f|^p d\mu \\
&\leq \int_{E_n} (2F)^p d\mu + \int_{E_n^c} \min\{\delta^p, (2F)^p\} d\mu \\
&< \frac{\epsilon}{2} + \int_\Omega h_\delta \, d\mu < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.
\end{aligned}$$

Thus, $\|f_n - f\|_p \to 0$. □

Problem 9. Show that polynomials are dense in $L_p([a, b])$.

Solution. We proceed in three steps using the triangle inequality of the norm $\|\cdot\|_p$. Let $f \in L_p([a, b])$ and $\epsilon > 0$.

1. **Finite-valued simple functions are dense in L_p:** First, we approximate f by a countable-valued simple function. Since f is measurable, there exists a sequence of countable-valued simple functions $\{\psi_k\}$ such that $|\psi_k| \nearrow |f|$ and $\psi_k \to f$ pointwise. Since $|\psi_k - f|^p \leq 2^p|f|^p \in L_1$, by the Lebesgue Convergence Theorem, $\|\psi_k - f\|_p \to 0$. We fix a ψ such that $\|f - \psi\|_p < \epsilon/4$.

   Next, let $\psi = \sum_{j=1}^{\infty} c_j \chi_{E_j}$ be this countable-valued simple function. Define the finite-valued partial sums $\phi_n = \sum_{j=1}^{n} c_j \chi_{E_j}$. Since $\psi \in L_p$, the series converges absolutely almost everywhere. We have $|\psi - \phi_n| \searrow 0$ pointwise as $n \to \infty$. By Levi Convergence Theorem applied to the sequence of functions $|\psi - \phi_n|^p$, we have $\|\psi - \phi_n\|_p \to 0$. Thus, there exists a finite-valued simple function ϕ such that $\|\psi - \phi\|_p < \epsilon/4$.

2. **Continuous functions are dense in finite-valued simple functions:** Let ϕ be the finite-valued simple function from the previous step. Since ϕ takes only finitely many values, it is bounded; let $M = \sup_{x\in[a,b]} |\phi(x)|$. By Lusin's Theorem, for any $\delta > 0$, there exists a function $g \in C([a,b])$ such that $\mu(\{x : \phi(x) \neq g(x)\}) < \delta$ and $\sup_{x\in[a,b]} |g(x)| \leq M$. We estimate the L_p distance:

$$\|\phi - g\|_p^p = \int_{[a,b]} |\phi - g|^p d\mu = \int_{\{\phi\neq g\}} |\phi - g|^p d\mu \leq \mu(\{\phi \neq g\}) \cdot (2M)^p < \delta(2M)^p.$$

   By choosing δ sufficiently small, we ensure $\|\phi - g\|_p < \epsilon/4$.

3. **Polynomials are dense in $C([a,b])$:** By the Weierstrass Approximation Theorem, for the continuous function g found above, there exists a polynomial P such that $\sup_{x\in[a,b]} |g(x) - P(x)| < \eta$. Then:

$$\|g - P\|_p = \left(\int_a^b |g(x) - P(x)|^p dx\right)^{1/p} \leq \eta(b-a)^{1/p}.$$

   Choosing η small enough yields $\|g - P\|_p < \epsilon/4$.

Combining these results via the triangle inequality:

$$\|f - P\|_p \leq \|f - \psi\|_p + \|\psi - \phi\|_p + \|\phi - g\|_p + \|g - P\|_p < \epsilon.$$

Thus, polynomials are dense in $L_p([a,b])$. □

Problem 10. Prove the reduction of the Lebesgue-Stieltjes integral $\int_{[a,b]} f dg = \int_{[a,b]} f g' d\mu$ for absolutely continuous g.

Solution. Since g is absolutely continuous, it is of bounded variation, so the Lebesgue-Stieltjes measure μ_g is well-defined and finite. Furthermore, g is differentiable almost everywhere and $g' \in L^1([a,b])$. We define the measure ν by $\nu(E) = \int_E g' d\mu$.

1. **Intervals:** Let $f = \chi_{[c,d]}$ where $[c,d] \subseteq [a,b]$. The LHS is $\int \chi_{[c,d]} dg = \mu_g([c,d]) = g(d) - g(c)$. The RHS is $\int_{[a,b]} \chi_{[c,d]} g' d\mu = \int_c^d g' d\mu$. Since g is absolutely continuous, the Fundamental Theorem of Lebesgue Calculus holds: $g(d) - g(c) = \int_c^d g'(t)dt$. Thus, the equality holds for intervals.

2. **Characteristic Functions of Measurable Sets:** Let $f = \chi_E$ for a Borel measurable set $E \subseteq [a,b]$. We have established in Step 1 that the two finite measures μ_g and ν agree on the algebra of finite unions of intervals. By the Uniqueness Theorem for Measures

(Carathéodory), they must agree on the σ-algebra generated by intervals (the Borel σ-algebra). Thus, $\mu_g(E) = \nu(E)$, which implies:

$$\int_{[a,b]} \chi_E dg = \int_{[a,b]} \chi_E g' d\mu.$$

(Note: For general Lebesgue measurable sets, the result follows by the completeness of the measures).

3. **Finite-valued Simple Functions:** Any finite-valued simple function ϕ can be written as $\phi = \sum_{k=1}^{n} c_k \chi_{E_k}$. By the linearity of the integral and the result from Step 2:

$$\int \phi dg = \sum_{k=1}^{n} c_k \int \chi_{E_k} dg = \sum_{k=1}^{n} c_k \int \chi_{E_k} g' d\mu = \int \phi g' d\mu.$$

4. **General Bounded Measurable Functions:** Let f be a bounded measurable function. We can decompose $f = f^+ - f^-$. Since f is bounded and the measures are finite, both parts are integrable. It suffices to prove the result for $f \geq 0$. There exists a sequence of non-negative, finite-valued simple functions $0 \leq \phi_n \nearrow f$. By the Levi Convergence Theorem applied to both measures μ_g and ν:

$$\int f dg = \lim_{n\to\infty} \int \phi_n dg = \lim_{n\to\infty} \int \phi_n g' d\mu = \int f g' d\mu.$$

Thus, the identity holds. $\square$